

Based on 48 sources

Universal Principle/Property	Self-Similarity Pattern (Equivariance)	Contradiction Gradient/Extreme	Algebraic Manifestation
Universality of Colimits	Invariance of colimit diagrams under pullback (f^*), particularly for Cartesian and coCartesian fibrations.	Paradoxical breakdown in non-right proper model categories where categorical fibrations fail to preserve limits under pullback.	Simplicial topology, Grothendieck and the structure
Kan Extension (Left/Right)	Free generation of functors from a small subcategory to a large category, preserving colimit structures.	Singularities at non-cofibrant objects where straightening/unstraightening procedures fail to yield strictly associative laws.	Stiefel-Whitney categories, correction C to L
Isometry and Equivalence	Equivalence between topological categories, simplicial categories, and quasi-categories via the simplicial nerve.	Torsion between strict and lax fiber products in higher categories ($n > 1$).	Nerve of categories, Cat _n and the realization
Self-Similarity in Ind-Objects	Ind-categories (Ind _κ) as formal filtered colimits preserving the core structural constants of small categories.	Entropy thresholds in infinite-dimensional spaces (e.g., Hilbert cube manifolds) where hypercompleteness fails.	Preshifts, P(C) embeddings
Geometric Descent (Effective Groupoids)	Every groupoid object in an ∞ -topos is effective, mirroring the surjectivity of maps in the category of sets.	Glitchons/Lacunons in non-effective groupoids where the Čech nerve fails to resolve the object.	Augmented objects, nerve categories